Nigerian Real Estate Market Analysis Dashboard

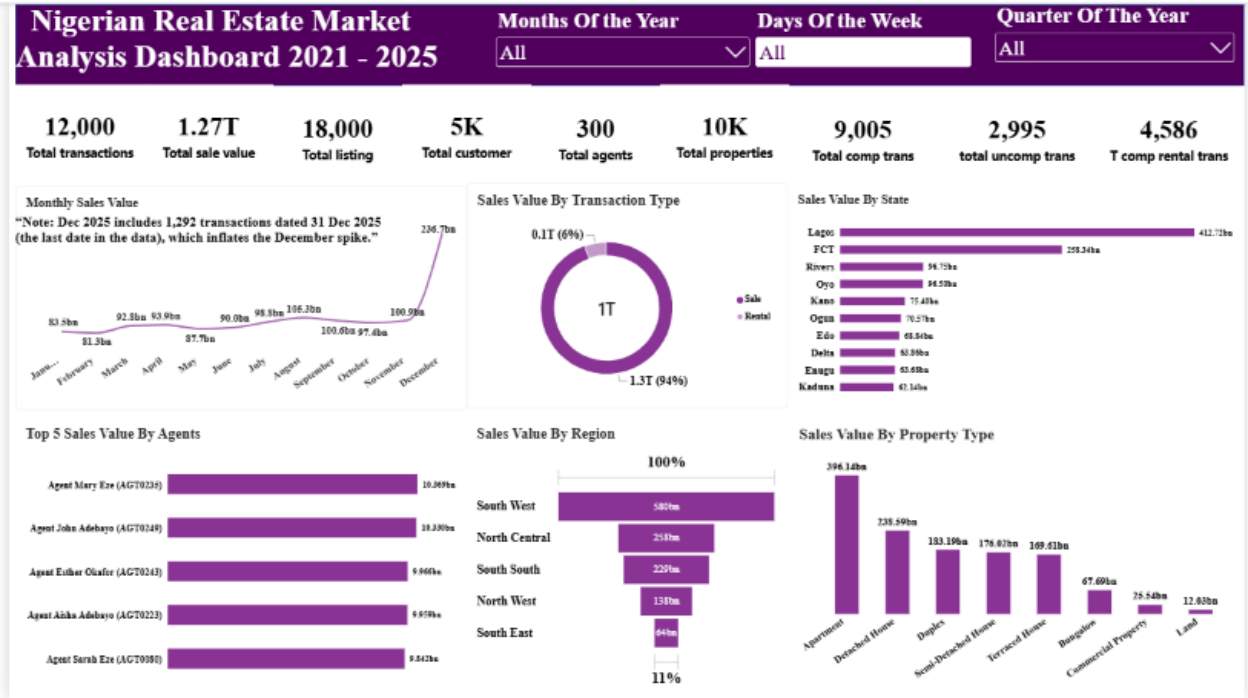
2021 - 2025

A Power BI data analysis project following a 15-step workflow

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Tools: Power BI (Power Query, data modelling, DAX, visuals)

Date: November 2026



1. Project Overview

This project turns raw Nigerian real estate data into business intelligence. A five-page Power BI dashboard shows how the market performs across time, locations, property types, customers and agents, so that a real estate company can decide where to focus, what to sell or rent, and who to target.

The project follows a 15-step data analysis workflow, from defining the business objective to monitoring:

1. Understand the business objective
2. Agree clear objectives
3. Define business and analytical questions
4. Data collection
5. Import data
6. Understand the data
7. Data cleaning and data quality
8. Data modelling
9. Feature engineering
10. EDA and key metrics (KPIs)
11. Visualize
12. Interpretation and communication
13. Action and recommendation
14. Documentation
15. Monitor

Project at a glance

Item	Detail
Data	7 CSV tables (2 fact tables, 5 dimension tables), covering 2021 to 2025
Records	12,000 transactions, 18,000 listings, 10,025 properties, 5,020 customers, 300 agents
Output	Five-page interactive Power BI dashboard
Headline result	1.27 trillion naira of completed sales value

2. Business Objective and Key Questions

Steps 1 to 3: business objective, clear objectives, analytical questions.

Business objective

To analyse Nigerian real estate listing and transaction data in order to understand how the market is performing, how business activity changes over time, and which locations, property types, customers and agents generate the most value. The analysis supports data-driven decisions on where to focus, what to sell or rent, who to target, and how to manage agent performance.

The overall question

"Where is the business activity, what is driving sales value, who should we target, and where should we focus?"

Analytical questions

Theme	Questions
Market performance	How many listings, transactions, properties, customers and agents are there? How many transactions are completed and how many cancelled?
Trends over time	How does sales value change by month, quarter and day of the week?
Sales and rentals	How is value split between sales and rentals, and how many rentals are completed?
Location performance	Which states, regions and cities generate the highest sales value?
Property performance	Which property types and furnishing statuses generate the most sales value?
Customer analysis	Which income segments, age groups, genders, occupations and marital statuses bring the most sales value?
Agent performance	Who are the top agents by sales value, and how does value differ by specialization?
Listing status	How are listings split by status and by type?

3. Data Collection and Understanding

Steps 4 and 6: data collection and understanding the data.

The dataset is a star schema of seven CSV files: two fact tables that record events, and five-dimension tables that describe them.

Table	Type	Rows	Description
fact listing	Fact	18,000	Property listings for sale or rent, with asking price and days on market
fact transaction	Fact	12,000	Sales and rentals, with amount, annual rent and commission
dim property	Dimension	10,025	Property type, bedrooms, bathrooms, size, year built, furnishing
dim customer	Dimension	5,020	Demographics, occupation, income segment, home location
dim agent	Dimension	300	Agent profile, specialization, years of experience
dim location	Dimension	34	City, state and region
dim date	Dimension	1,826	Calendar table covering 2021 to 2025

Data dictionary

`dim location`

Field	Type	Key	Description
`location_id`	Text	PK	Unique identifier for each location
`city`	Text	-	City or locality where the property is located
`state`	Text	-	Nigerian states
`region`	Text	-	Broad Nigerian geographic region
`active location`	Integer	-	Whether the location is active in the dataset

`dim date`

Field	Type	Key	Description
`date_id`	Integer	PK	Date in YYYYMMDD format
`date`	Date	-	Calendar date
`year`	Integer	-	Calendar year
`quarter`	Text	-	Quarter of the year (Q1 to Q4)
`month number`	Integer	-	Month number, 1 to 12
`month name`	Text	-	Name of the month
`month year`	Text	-	Month and year, e.g. Jan-2024
`week number`	Integer	-	ISO week number
`day`	Integer	-	Day of the month
`day name`	Text	-	Name of the day
`is weekend`	Boolean	-	Whether the date falls on a weekend

`dim property`

Field	Type	Key	Description
`property_id`	Text	PK	Unique identifier for each property

Field	Type	Key	Description
location_id	Text	FK	Location of the property
property type	Text	-	Type of property
bedrooms	Integer	-	Number of bedrooms
bathrooms	Integer	-	Number of bathrooms
property_size_sqm	Decimal	-	Size in square meters
year built	Integer	-	Year of construction
furnishing status	Text	-	Furnishing condition

dim customer

Field	Type	Key	Description
customer	Text	PK	Unique identifier for each customer
first name	Text	-	First name
last name	Text	-	Surname
gender	Text	-	Gender
age	Integer	-	Age
marital status	Text	-	Marital status
occupation	Text	-	Occupation
income segment	Text	-	Broad income classification
home_location_id	Text	FK	Location of the customer's home

dim agent

Field	Type	Key	Description
agent_id	Text	PK	Unique identifier for each agent
agent name	Text	-	Name assigned to the agent
gender	Text	-	Gender
age	Integer	-	Age
base_location_id	Text	FK	Agent's primary operating location
specialization	Text	-	Primary area of specialization
years experience	Integer	-	Years in real estate

fact listing

Field	Type	Key	Description
listing_id	Text	PK	Unique identifier for each listing
property_id	Text	FK	Property listed
agent_id	Text	FK	Agent responsible
date_id	Integer	FK	Link to the date table
listing date	Date	-	Date the property was listed
listing type	Text	-	For sale or for rent

Field	Type	Key	Description
listing status	Text	-	Status of the listing
asking price	Decimal	-	Price requested
days_on_market	Integer	-	Days the property stayed on the market

fact transaction

Field	Type	Key	Description
transaction_id	Text	PK	Unique identifier for each transaction
listing_id	Text	FK	Listing behind the transaction
property_id	Text	FK	Property involved
customer_id	Text	FK	Customer involved
agent_id	Text	FK	Agent responsible
date_id	Integer	FK	Link to the date table
transaction date	Date	-	Date of the transaction
transaction type	Text	-	Sale or rental
transaction status	Text	-	Completed or cancelled
transaction amount	Decimal	-	Financial value of the transaction
annual rent	Decimal	-	Annual rent (rental transactions)
commission_rate	Decimal	-	Commission percentage
commission_amount	Decimal	-	Commission value

What the data checks showed

- Every table has a unique primary key and the key columns have no empty values.
- Agent names repeat: the 300 agents share only 184 distinct names. The agents are different people (they differ in age, gender, location and experience), so agent_id is the true identifier.
- 121 properties (1.2%) have no recorded size, 80 have no furnishing status, and 100 customers have no occupation.
- 1,292 transactions (10.8%) are dated 31 December 2025, which is the last date in the data.

4. Data Import and Cleaning

Steps 5 and 7: import data, data cleaning and data quality.

All seven CSV files were loaded into Power BI with Get Data and cleaned in Power Query. Each table has the same base steps; two tables have an extra step.

Table	Applied steps
dim agent	Source > Promoted Headers > Changed Type > Removed Duplicates (agent_id)
dim customer	Source > Promoted Headers > Changed Type > Removed Duplicates > Replaced Value (blank occupation to "Not Specified")

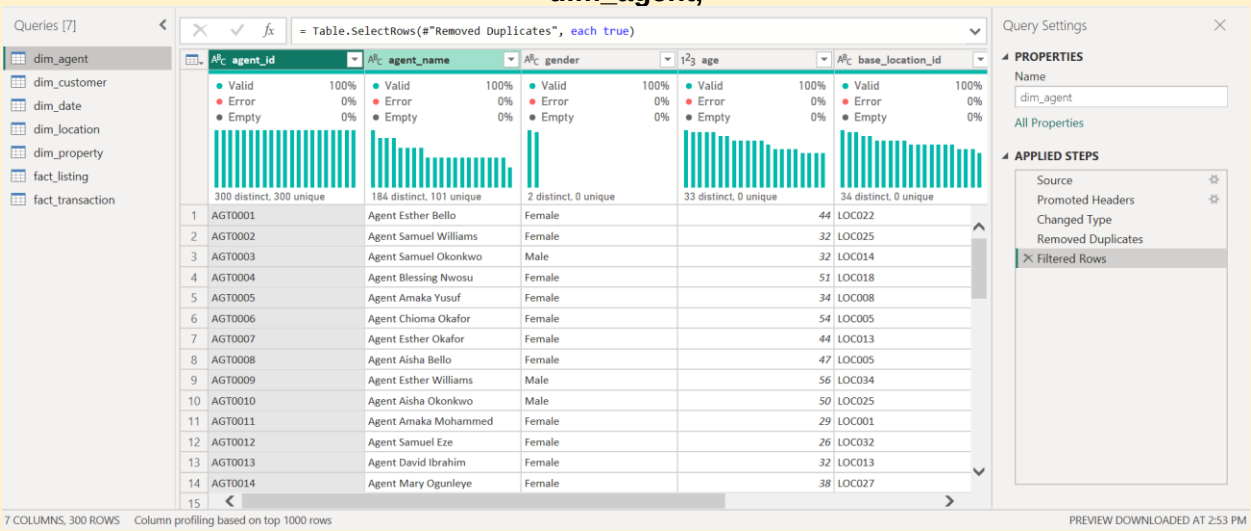
Table	Applied steps
dim date	Source > Promoted Headers > Changed Type > Removed Duplicates (date_id)
dim location	Source > Promoted Headers > Changed Type > Removed Duplicates (location_id)
dim property	Source > Promoted Headers > Changed Type > Removed Duplicates > Replaced Value (blank furnishing status to "Not Specified")
fact listing	Source > Promoted Headers > Changed Type > Removed Duplicates (listing_id)
fact transaction	Source > Promoted Headers > Changed Type > Removed Duplicates (transaction_id)

Data quality decisions

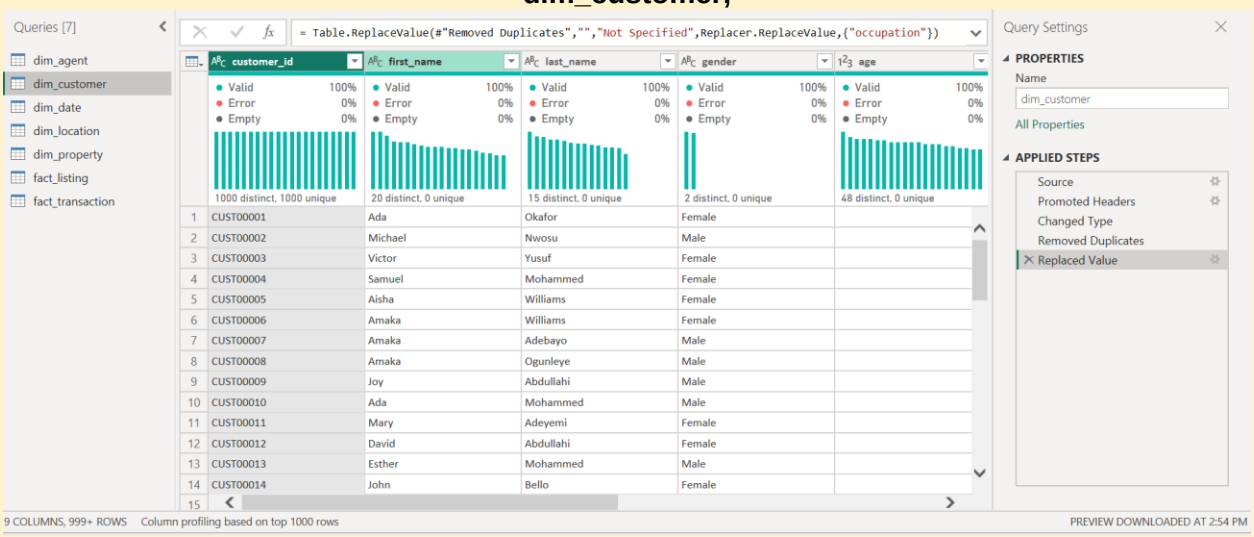
- Headers were promoted and data types set on every table.
- Duplicates were removed on each table's primary key. In the tables I checked, the row counts did not change, so this step acted as a safeguard.
- Blank text values were replaced with "Not Specified": 80 in furnishing_status and 100 in occupation. They now appear as a category in charts instead of blanks.
- The 121 blank property_size_sqm values were left null on purpose. Filling them with an average would invent sizes, and they are excluded from price-per-sqm calculations. They are spread across all property types, so they are random gaps, not a special kind of property.

Power Query screenshot for each of the 7 tables showing the Applied Steps pane (

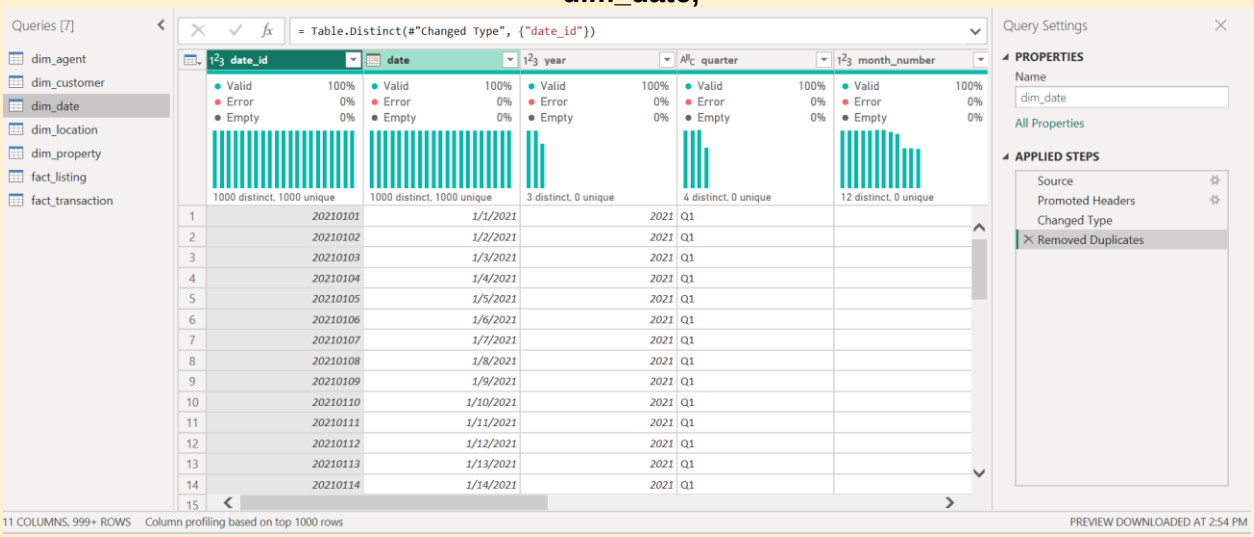
dim_agent,



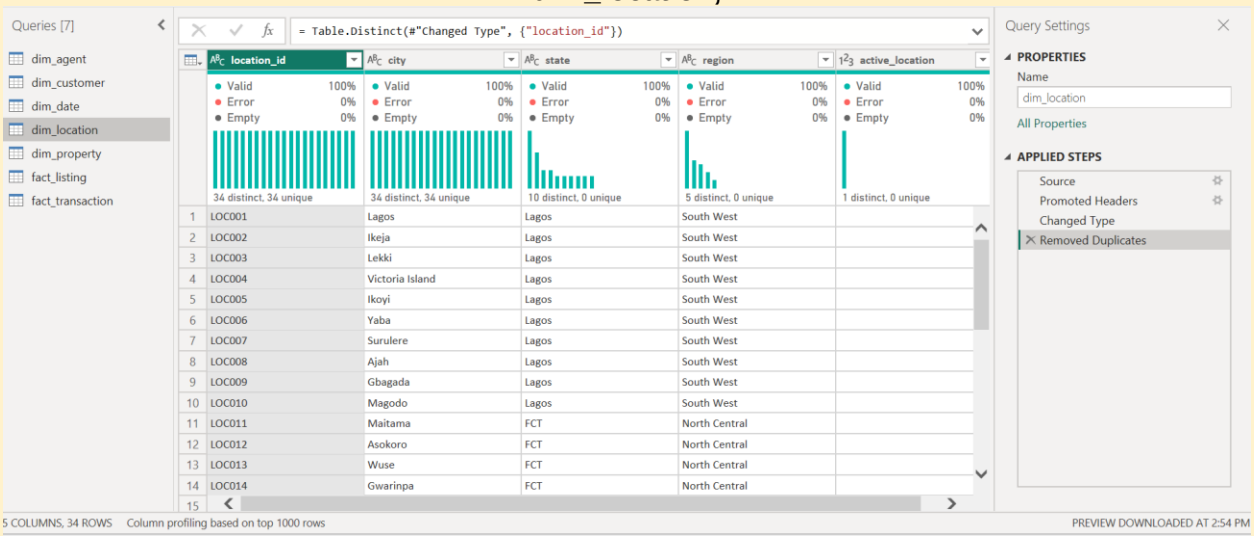
dim_customer,



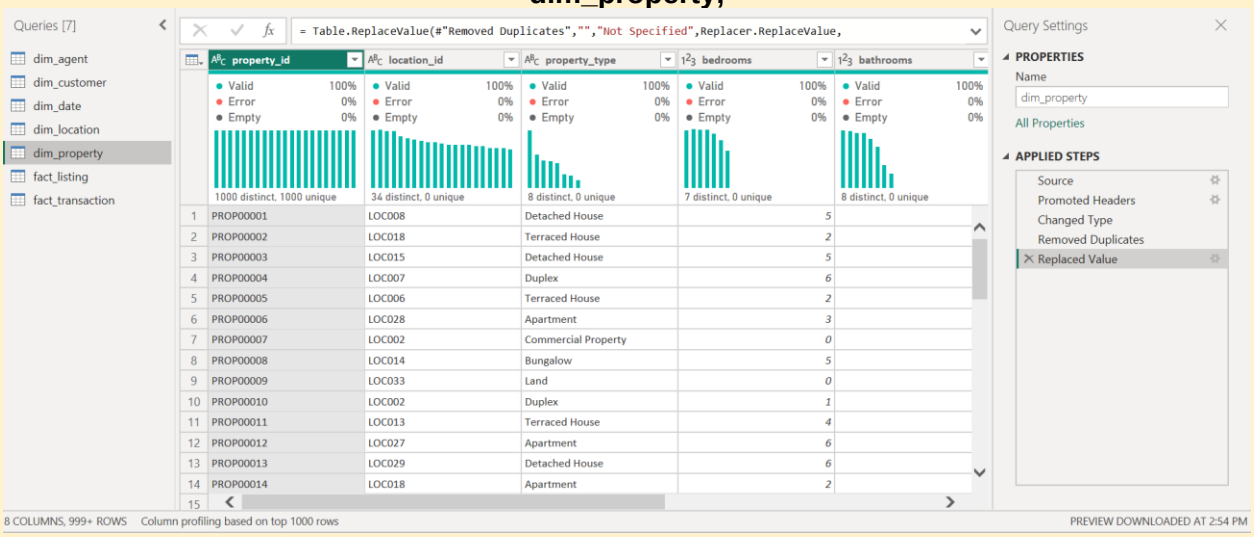
dim_date,



dim_location,

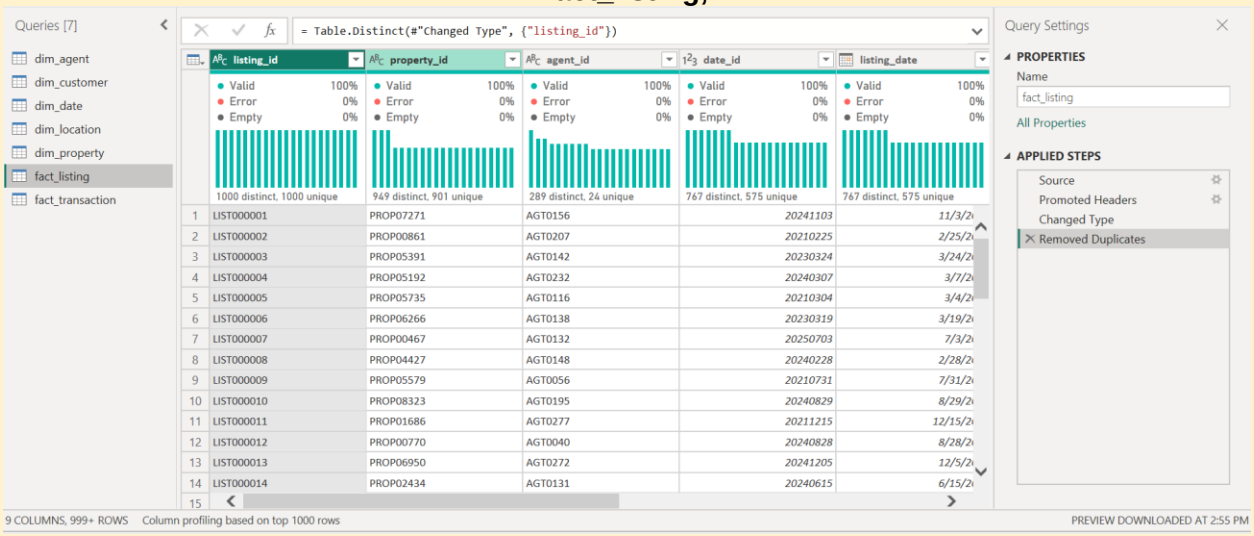


dim_property,



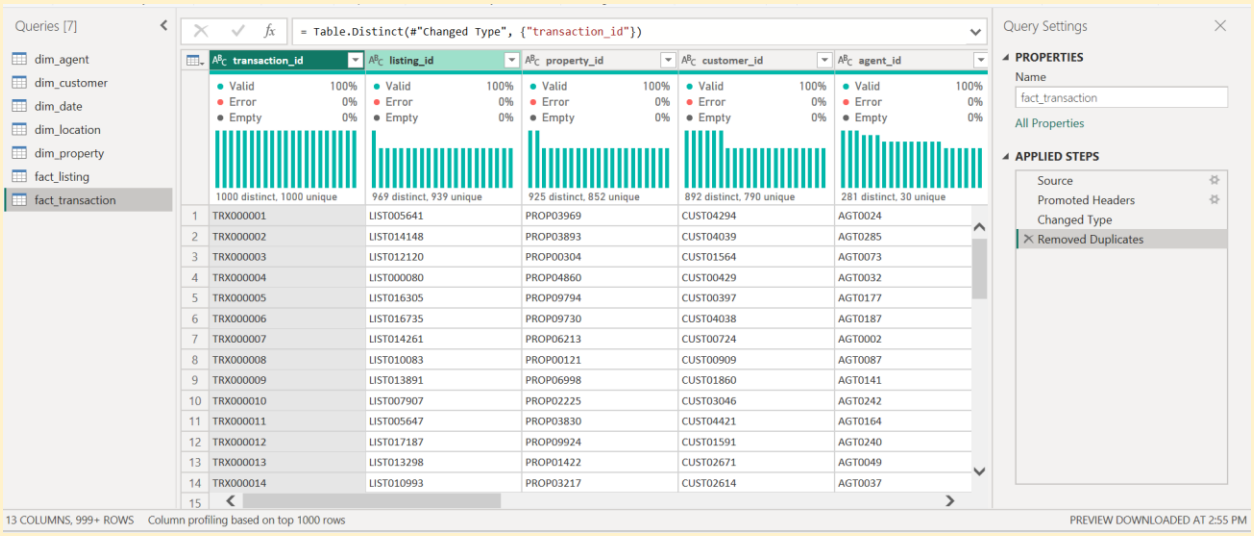
	property_id	location_id	property_type	bedrooms	bathrooms
1	PROP00001	LOC008	Detached House	5	
2	PROP00002	LOC018	Terraced House	2	
3	PROP00003	LOC015	Detached House	5	
4	PROP00004	LOC007	Duplex	6	
5	PROP00005	LOC006	Terraced House	2	
6	PROP00006	LOC028	Apartment	3	
7	PROP00007	LOC002	Commercial Property	0	
8	PROP00008	LOC014	Bungalow	5	
9	PROP00009	LOC033	Land	0	
10	PROP00010	LOC002	Duplex	1	
11	PROP00011	LOC013	Terraced House	4	
12	PROP00012	LOC027	Apartment	6	
13	PROP00013	LOC029	Detached House	6	
14	PROP00014	LOC018	Apartment	2	

fact_listing,



	listing_id	property_id	agent_id	date_id	listing_date
1	LIST000001	PROP07271	AGT0156	20241103	11/3/2(
2	LIST000002	PROP00861	AGT0207	20210225	2/25/2(
3	LIST000003	PROP05391	AGT0142	20230324	3/24/2(
4	LIST000004	PROP05192	AGT0232	20240307	3/7/2(
5	LIST000005	PROP05735	AGT0116	20210304	3/4/2(
6	LIST000006	PROP06266	AGT0138	20230319	3/19/2(
7	LIST000007	PROP00467	AGT0132	20250703	7/3/2(
8	LIST000008	PROP04427	AGT0148	20240228	2/28/2(
9	LIST000009	PROP05579	AGT0056	20210731	7/31/2(
10	LIST000010	PROP08323	AGT0195	20240829	8/29/2(
11	LIST000011	PROP01686	AGT0277	20211215	12/15/2(
12	LIST000012	PROP00770	AGT0040	20240828	8/28/2(
13	LIST000013	PROP06950	AGT0272	20241205	12/5/2(
14	LIST000014	PROP02434	AGT0131	20240615	6/15/2(

fact_transaction)



5. Data Modelling

Step 8: tables, relationships, keys, fact and dimension tables.

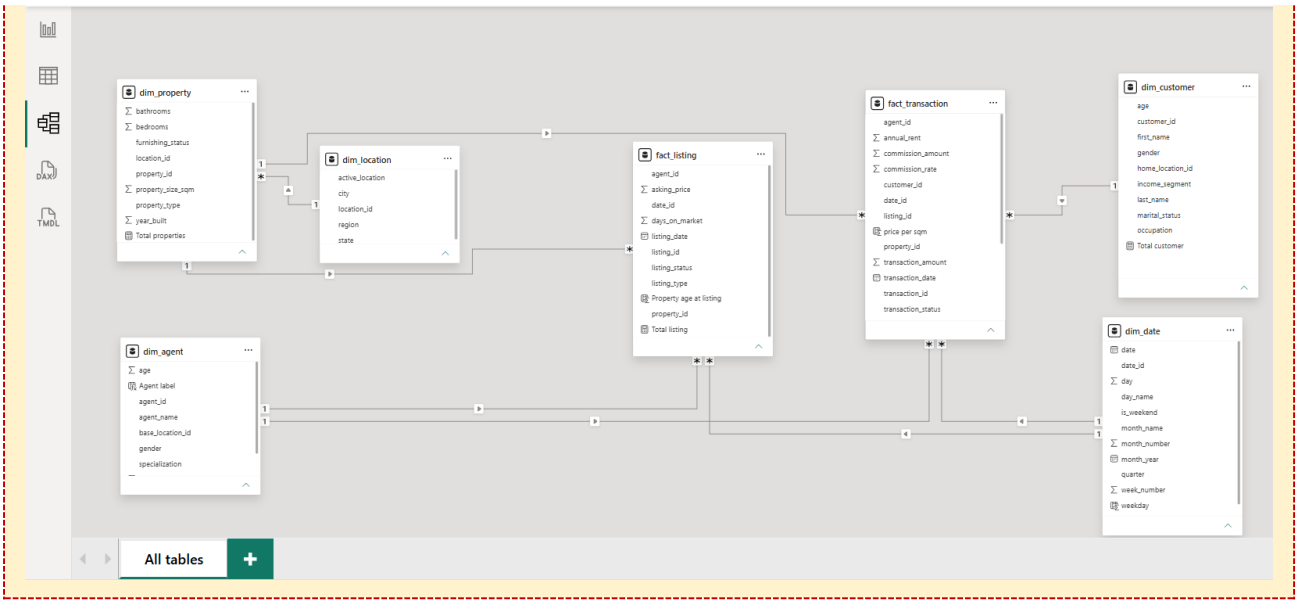
The model is a star schema. The two fact tables (fact listing and fact transaction) are surrounded by five dimension tables. All relationships are one-to-many, with the "one" side on the dimension.

Dimension (one)	Fact or table (many)	Key
dim location	dim property	location_id
dim property	fact listing, fact transaction	property_id
dim agent	fact listing, fact transaction	agent_id
dim date	fact listing, fact transaction	date_id
dim customer	fact transaction	customer_id

Modelling decisions

- dim customer[home_location_id] (where a customer lives) and dim agent[base_location_id] (where an agent works) are not linked to dim location. They describe different things from a property's location, and linking them would create ambiguous relationships.
- fact listing and fact transaction are not linked to each other. Both connect through the shared dimensions. As a result, listing fields cannot filter transaction values, so the listing charts use counts of listings.

INSERT SCREENSHOT HERE: Model view screenshot showing all 7 tables and their relationships



6. Feature Engineering

Step 9: calculated columns, ratios, date parts, labels and DAX measures.

Feature engineering added 4 calculated columns (row-level fields) and 11 DAX measures (calculations that respond to filters and slicers).

Calculated columns

Property age at listing (fact listing) Age of the property when it was listed

```
Property age at listing = YEAR(fact_listing[listing_date].[Date]) -
RELATED(dim_property[year_built])
```

price per sqm (fact transaction) Transaction value per square metre

```
price per sqm = DIVIDE(fact_transaction[transaction_amount],
RELATED(dim_property[property_size_sqm]))
```

weekday (dim date) Day-of-week number

```
weekday = WEEKDAY(dim_date[date].[Date])
```

Agent label (dim agent) Unique label for each agent, because names repeat

```
Agent label = dim_agent[agent_name] & " (" & dim_agent[agent_id] & ")"
```

Measures

Total properties (dim property) Number of properties

```
Total properties = DISTINCTCOUNT(dim_property[property_id])
```

Total listing (fact listing) Number of listings

```
Total listing = DISTINCTCOUNT(fact_listing[listing_id])
```

Total customer (dim customer) Customers on record

```
Total customer = COUNT(dim_customer[customer_id])
```

Total agents (dim agent) Number of agents

```
Total agents = COUNT(dim_agent[agent_id])
```

Total transactions (fact transaction) Number of transactions

```
Total transactions = COUNT(fact_transaction[agent_id])
```

Total comp trans (fact transaction) Completed transactions

```
Total comp trans = CALCULATE([Total transactions], fact_transaction[transaction_status] =
"Completed")
```

total uncomp trans (fact transaction) Cancelled transactions

```
total uncomp trans = CALCULATE([Total transactions], fact_transaction[transaction_status] =
"Cancelled")
```

T comp rental trans (fact transaction) Completed rentals

```
T comp rental trans = CALCULATE(COUNT(fact_transaction[transaction_id]),
fact_transaction[transaction_status] = "Completed", fact_transaction[transaction_type] =
"Rental")
```

total completed sales transactions (fact transaction) Completed sales

```
total completed sales transactions = CALCULATE(COUNT(fact_transaction[transaction_id]),
fact_transaction[transaction_status] = "Completed", fact_transaction[transaction_type] =
"Sale")
```

Total sale value (fact transaction) Value of completed sales only

```
Total sale value = CALCULATE(SUM(fact_transaction[transaction_amount]),
fact_transaction[transaction_status] = "Completed", fact_transaction[transaction_type] =
"Sale")
```

Completed transaction value (fact transaction) Value of all completed transactions, sales and rentals

```
Completed transaction value = CALCULATE(SUM(fact_transaction[transaction_amount]),
fact_transaction[transaction_status] = "Completed")
```

Design note

Total sale value filters on transaction type inside the measure. When it was placed next to transaction type in a donut chart, that inner filter overrode the chart's own Rental/Sale split, and both slices showed the same value. The Completed transaction value measure has no type filter, so it is the one used in the donut chart, which then shows the correct 94% sales and 6% rentals split.

7. EDA and Key Metrics (KPIs)

Step 10: exploring the data and defining the KPIs that matter.

KPI	Value	Definition
Total transactions	12,000	All transactions, completed and cancelled
Total sale value	1.27T	Value of completed sales only
Total listing	18,000	Distinct listings
Total customer	5K (5,020)	Customers on record
Total agents	300	Distinct agents
Total properties	10K (10,025)	Distinct properties
Total comp trans	9,005	Completed transactions
Total uncomp trans	2,995	Cancelled transactions

KPI	Value	Definition
T comp rental trans	4,586	Completed rental transactions

Further numbers from exploring the raw data:

- Completed sales: 4,419 transactions worth about 1.269T. Average completed sale: about 287 million naira.
- Completed rentals: 4,586 transactions worth about 75 billion naira. Average completed rental: about 16 million naira.
- Cancellation rate: 24.96% of all transactions, with about 426 billion naira of sale value cancelled. The rate is almost the same for sales (25.4%) and rentals (24.5%), in every year, and across states (22.6% to 26.0%) and property types (23.6% to 26.7%). Across individual agents with at least 30 transactions it ranges from 9% to 46%.
- Listings: 6,316 active, 5,302 sold, 3,310 rented, 1,812 expired and 1,260 withdrawn. For sale and for rent are split almost evenly (9,003 and 8,997).

The cancellation figures were calculated from the raw data. The dashboard shows only the counts of completed and cancelled transactions.

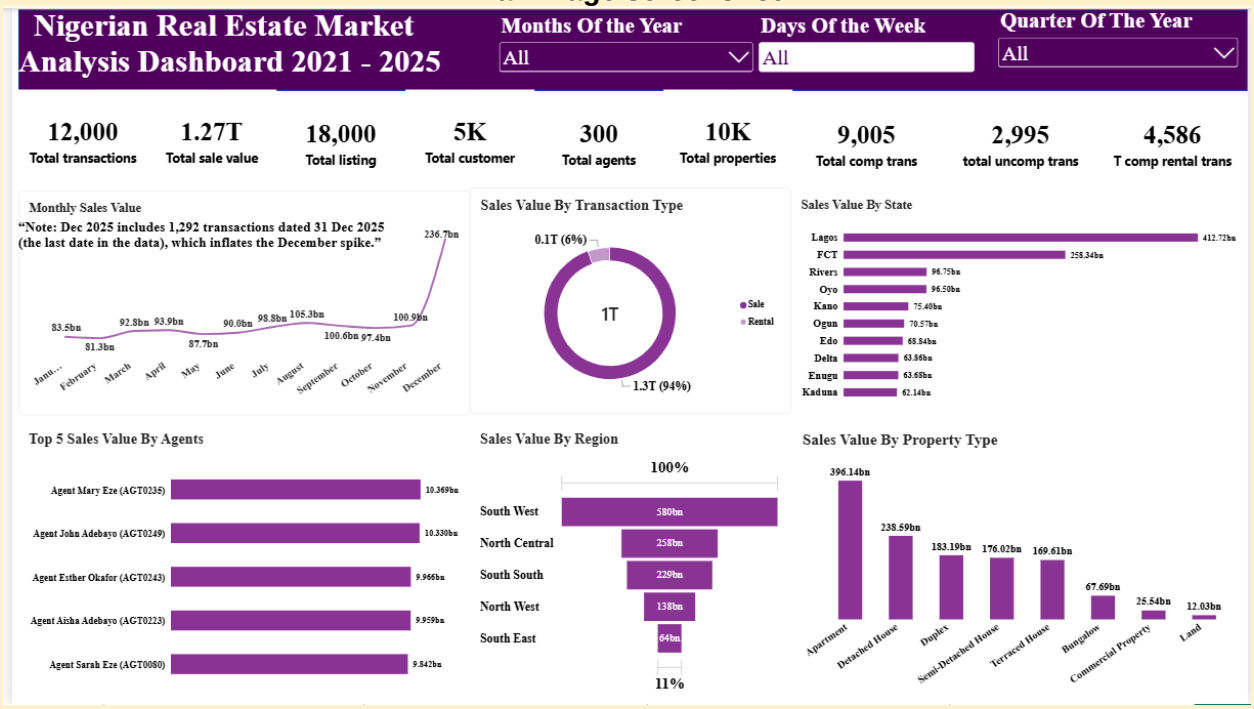
8. Visualization: The Dashboard

Step 11: charts and dashboards that suit the audience. The dashboard has five pages. The Main Page has slicers for month of the year, day of the week and quarter of the year.

Page 1: Main Page

Nine KPI cards (total transactions, total sale value, total listing, total customer, total agents, total properties, total completed, cancelled and completed rental transactions), plus monthly sales value, sales value by transaction type (donut), by state, by region (funnel) and by property type, and the top 5 agents by sales value (filtered to the top 5 using the Agent label). A text note on the monthly chart explains the 31 December 2025 effect.

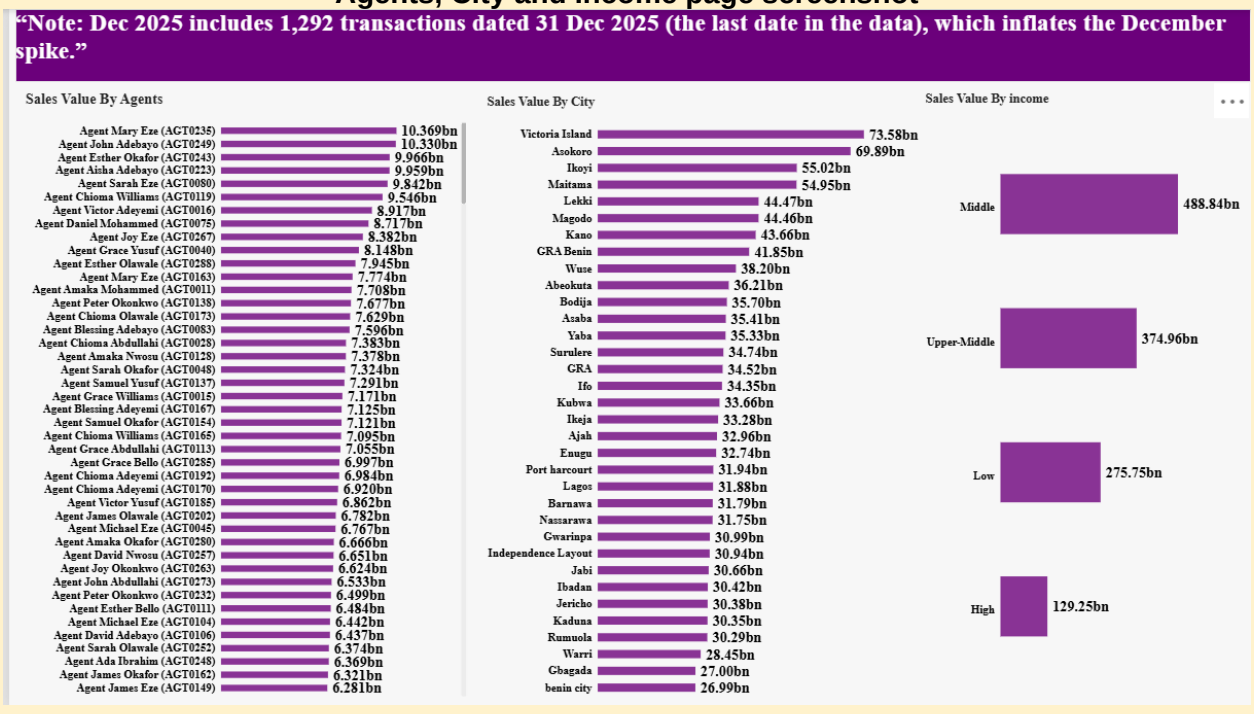
Main Page screenshot



Page 2: Agents, City and Income

Sales value by agent (using the unique agent label), by city and by customer income segment.

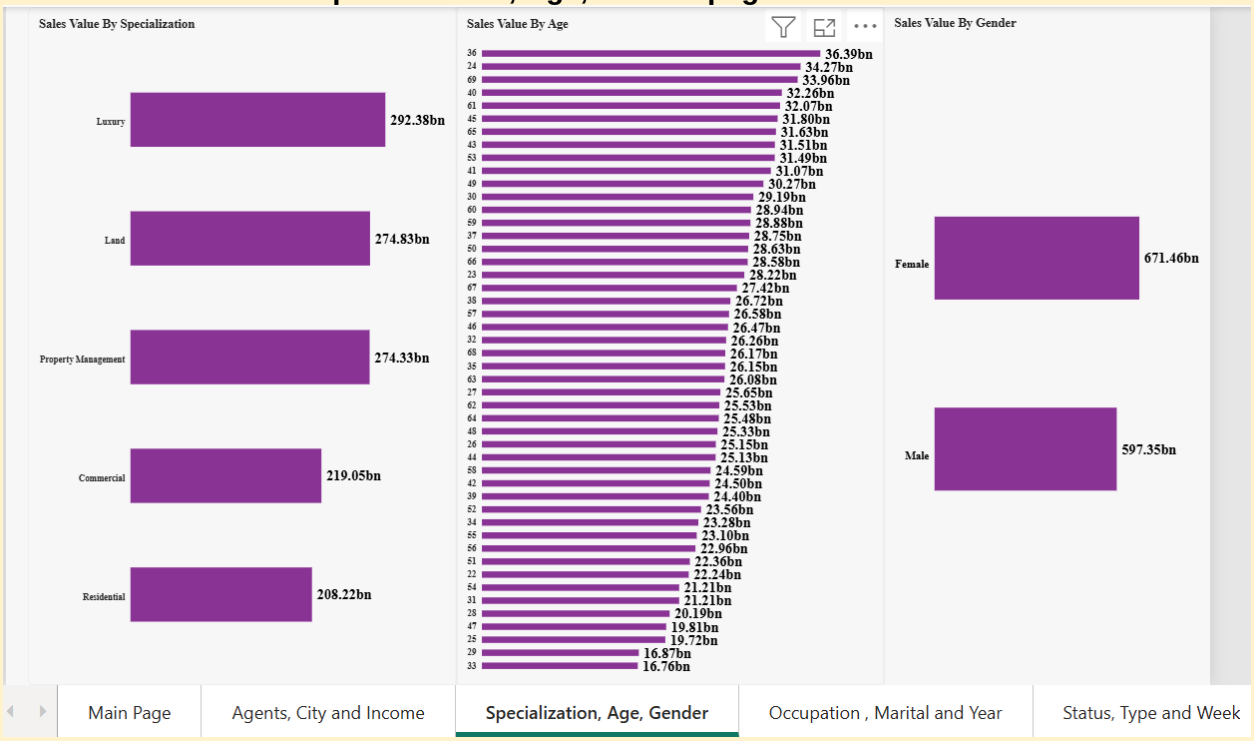
Agents, City and Income page screenshot



Page 3: Specialization, Age, Gender

Sales value by agent specialization, and by customer age and gender.

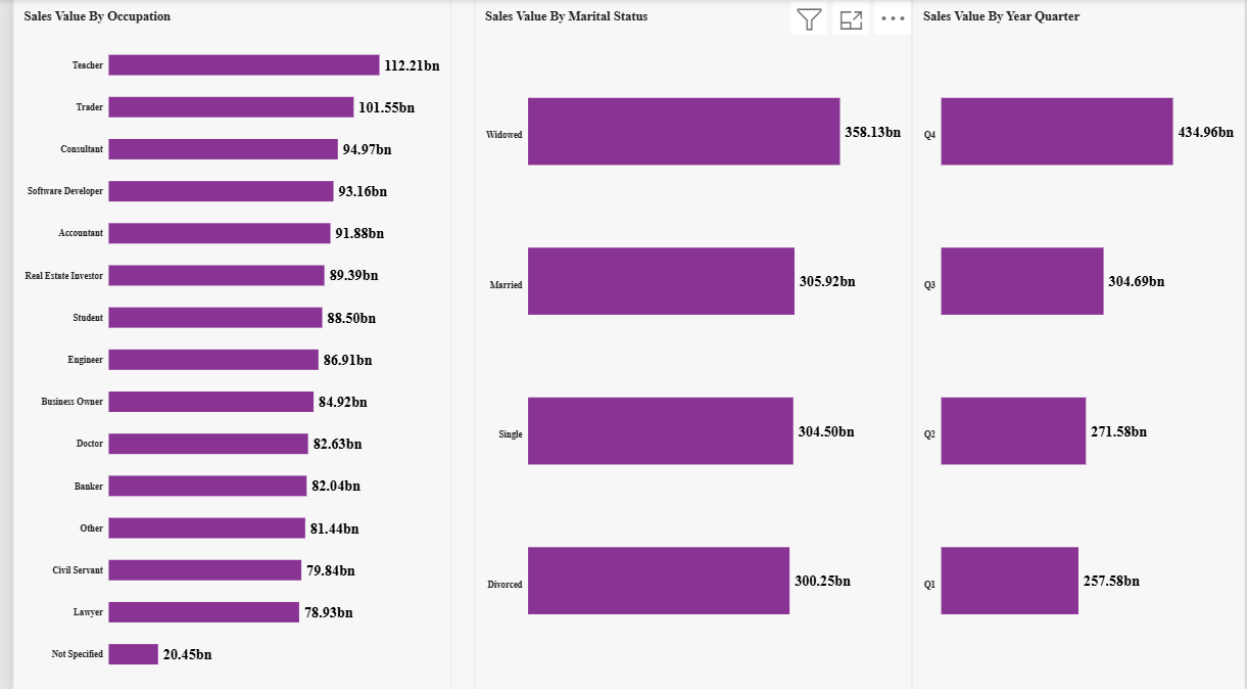
Specialization, Age, Gender page screenshot



Page 4: Occupation, Marital and Year

Sales value by customer occupation and marital status, and by quarter of the year.

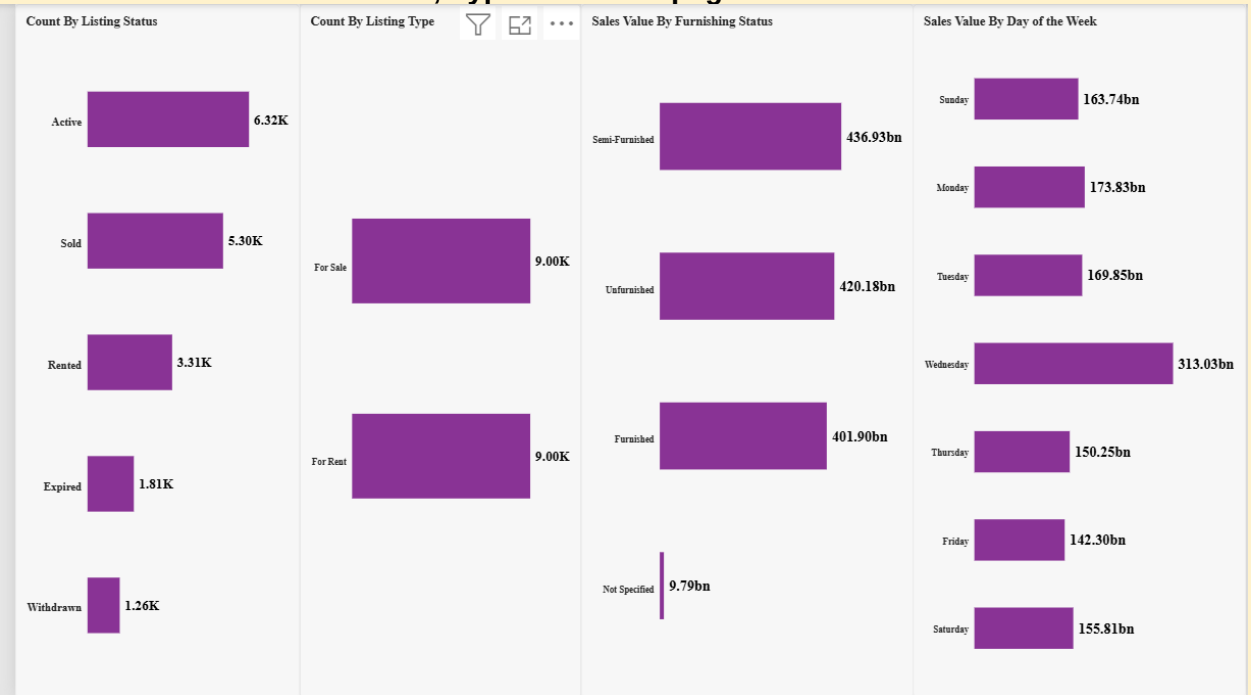
Occupation, Marital and Year page screenshot



Page 5: Status, Type and Week

Count of listings by status and by type, and sales value by furnishing status and by day of the week.

Status, Type and Week page screenshot



9. Interpretation and Communication

Step 12: what the numbers mean, why they happened, and how they link back to the problem.

16. **Value is concentrated in two states.** Lagos (about 413bn) and FCT (about 258bn) give 53% of all sales value. By region, South West leads with about 580bn (46%), then North Central (258bn), South South (229bn), North West (138bn) and South East (64bn).
17. **Premium neighbourhoods lead the cities.** Victoria Island (73.6bn), Asokoro (69.9bn), Ikoyi (55.0bn) and Maitama (55.0bn) are the top four cities.
18. **Apartments and Detached Houses make up half of the value.** Apartments bring about 396bn (31%) and Detached Houses about 239bn. Land (12bn) and Commercial Property (25.5bn) together are only about 3%.
19. **Sales carry the value; rentals carry the volume.** Sales are 94% of completed value (about 1.27T) and rentals 6% (about 75bn). By count they are almost equal (4,419 sales and 4,586 rentals) because the average sale is about 287 million naira against about 16 million for a rental.
20. **The market grew steadily through the year.** Quarterly sales value rises from about 258bn in Q1 to about 272bn in Q2 and about 305bn in Q3. The large Q4 bar (435bn) and the Wednesday peak (313bn) are caused by the 31 December 2025 cluster (see the data caveat below). Without it, Q4 is about 300bn and Wednesday about 178bn. Monday is the strongest day (174bn) and Friday the weakest (142bn).
21. **Sales are spread very evenly across the 300 agents.** The top five individual agents (Mary Eze, John Adebayo, Esther Okafor, Aisha Adebayo and Sarah Eze) each bring about 10bn, only about 4% of sales value in total. Agent specializations are balanced too, from Residential (208bn) to Luxury (292bn).
22. **Middle and upper-middle income customers bring most of the value.** Middle (489bn) and Upper-Middle (375bn) together give 68% of sales value. Low income brings 276bn and High income 129bn (10%).
23. **Gender, age, occupation and marital status are weak separators.** Female customers account for 671bn (53%) and male customers for 597bn (47%). No single age dominates. Occupations range from Teacher (112bn) to Lawyer (79bn), and marital statuses from 300bn to 358bn.
24. **Furnishing status makes little difference.** Semi-Furnished (437bn), Unfurnished (420bn) and Furnished (402bn) are close.
25. **Cancellations are the main risk.** About 1 in 4 transactions is cancelled, which put about 426bn of sale value at risk. Because the rate is almost the same in every state, property type and year, it points to the sales process, not to one place or product.

Data caveat: 31 December 2025

1,292 transactions (10.8% of all) are dated 31 December 2025, the last date in the data. A normal day has about 7. About 135bn of completed sales fall on that one day. This inflates December, Q4, Wednesday and the 2025 totals. The most likely cause is dates clipped to the end of the data, because 71% of these transactions would have closed after that date. The data was left unchanged and the dashboard carries a note.

10. Actions and Recommendations

Step 13: practical next steps for the business.

26. **Concentrate on Lagos, FCT and the premium neighbourhoods.** Lagos and FCT give 53% of sales value, and Victoria Island, Asokoro, Ikoyi and Maitama lead the cities. Also investigate the South East (about 5% of value) to find out whether it is an untapped market or a gap in presence.
27. **Prioritise Apartments and Detached Houses.** They are half of sales value. Review whether Land and Commercial Property deserve the same effort, since they are about 3%.
28. **Treat sales as the revenue engine and rentals as a volume line.** The average rental is a small fraction of the average sale, so rental pricing and commission are worth reviewing.
29. **Target middle and upper-middle income customers.** They bring 68% of value. High income is only 10%, so it is either a niche or an opportunity for luxury offers. Gender, age, occupation and marital status are poor targeting criteria because value is spread evenly across them.
30. **Reduce cancellations, the biggest risk.** Each 1-point drop in the cancellation rate is worth roughly 17bn of sale value. Review the sales process (deposits, buyer qualification and follow-up), and start with the agents whose cancellation rates are highest.
31. **Lift the whole team and learn from the top agents.** Because sales do not depend on a few agents, gains will come from raising the team. Study what the best-performing agents do differently and share it.
32. **Fix date capture at the source.** Check the transactions dated 31 December 2025 with the owner of the data, so that trend analysis becomes reliable.

11. Documentation

Step 14: method, assumptions and data sources.

Tools used

Power BI Desktop (Power Query for cleaning, Model view, DAX for measures and calculated columns, and report visuals), 7 CSV source files, Word and PDF for documentation, and GitHub for publishing.

Facts

- "Sales value" means the value of completed sales only. Cancelled transactions and rentals are excluded from Total sale value.
- For rentals, transaction_amount is the annual rent.
- State, region and city come from the property's location.
- Month, quarter and day-of-week charts combine all five years (2021 to 2025).
- Agents are identified by agent_id, because agent names repeat.

Limitations

- 31 December 2025 cluster: 1,292 transactions (about 135bn of completed sales) share one date, inflating December, Q4, Wednesday and 2025. The data was left as it is and a note was added to the dashboard.
- 121 properties (1.2%) have no size. They were left null and are excluded from price-per-sqm calculations.

- The price per sqm column mixes sale prices and annual rents. It is not shown on the current pages and, if used later, should be limited to sales.
- Blank values were replaced with "Not Specified": 80 in furnishing status and 100 in occupation.
- "Total customer" counts all customers on record, including those with no transaction.
- fact_listing and fact_transaction are not linked, so listing fields cannot filter transaction values. The listing charts use counts.
- Agent names repeat (300 agents share 184 names), so charts use a unique label combining name and ID.
- The cancellation rate and cancelled value were calculated from the raw data and are not shown as a chart.

12. Monitoring

Step 15: tracking results and repeating the cycle when needed.

To keep the dashboard current, replace or refresh the source files and click Refresh in Power BI. Review these KPIs regularly: total sale value, completed against cancelled transactions, and listings by status. After each refresh, check again for clusters of transactions on a single date. Rerun the full cycle if the business goals or the structure of the data change.